Table 1: C3 per system, sequence OOD (dev). BL = HyperKKL ($r = 4$; HKKL), BF = best model-based filter, BW = best WORKING model-based filter (below climatology; $k/n^{\dagger}$ = failed / present), DD = best data-driven same-data baseline. Changes: paired relative change of BL ± 95% CI half-width (Student-t over seeds; bold/underline: significant). FAILED filters are never used as the reference. Abbreviations: -or oracle, -Hu Huber, GB grey-box, Koopman-bi bilinear Koopman KF.

scenario / plant (clim.)	BL	NRMSE	BF	NRMSE	BW	NRMSE	fail.	(a) BL vs BW	DD	NRMSE	(b) BL vs DD
exact											
battery (0.78)	HyperKKL	0.0738	EKF	0.00521	EKF	0.00521	1/8	+1317% ±363	pend. 0/5		–
bioreactor (0.928)	HyperKKL	0.0661	EKF	0.00619	EKF	0.00619	0/8	+967% ±144	pend. 0/5		–
duffing (1.69)	HyperKKL	0.0547	UKF	0.00488	UKF	0.00488	0/8	+1021% ±88	pend. 0/5		–
lorenz (1)	HyperKKL	0.0259	UKF	0.00645	UKF	0.00645	0/8	+301% ±91	pend. 0/5		–
quad_tank (0.651)	HyperKKL	0.0712	UKF	0.0141	UKF	0.0141	0/8	+407% ±31	pend. 0/5		–
vdp (1.12)	HyperKKL	0.0189	EKF	0.00578	EKF	0.00578	0/8	+227% ±10	pend. 0/5		–
exact+outliers											
battery (0.78)	HyperKKL	2.13	PF	0.0293	PF	0.0293	4/8	+7168% ±3883	pend. 0/5		–
bioreactor (0.928)	HyperKKL	0.104	UKF-Hu	0.00619	UKF-Hu	0.00619	1/8	+1582% ±495	pend. 0/5		–
duffing (1.69)	HyperKKL	2.3	UKF-Hu	0.00481	UKF-Hu	0.00481	1/8	+47783% ±6123	pend. 0/5		–
lorenz (1)	HyperKKL	0.599	UKF-Hu	0.00693	UKF-Hu	0.00693	3/8	+8543% ±220	pend. 0/5		–
quad_tank (0.651)	HyperKKL	1.49	EKF-or	0.0252	EKF-or	0.0252	5/8	+5807% ±237	pend. 0/5		–
vdp (1.12)	HyperKKL	0.84	UKF-Hu	0.0059	UKF-Hu	0.0059	1/8	+14138% ±743	pend. 0/5		–
colored_t											
battery (0.78)	HyperKKL	0.0595	EKF-or	0.00989	EKF-or	0.00989	1/8	+502% ±94	GB-EKF-or	0.00989	+502% ±95
bioreactor (0.928)	HyperKKL	0.0645	EKF-or	0.00624	EKF-or	0.00624	0/8	+933% ±252	GB-EKF-or	0.00625	+932% ±251
duffing (1.69)	HyperKKL	0.0994	EKF-or	0.0104	EKF-or	0.0104	0/8	+860% ±45	GB-EKF-or	0.0104	+858% ±50
lorenz (1)	HyperKKL	0.0406	UKF-or	0.0177	UKF-or	0.0177	0/8	+129% ±8	GB-EKF-or	0.0183	+122% ±8
quad_tank (0.651)	HyperKKL	0.0696	EKF-or	0.0169	EKF-or	0.0169	0/8	+313% ±21	GB-EKF-or	0.017	+310% ±16
vdp (1.12)	HyperKKL	0.035	PF	0.0134	PF	0.0134	0/8	+161% ±18	GB-EKF-or	0.0137	+156% ±12
mismatch10											
battery (0.776)	HyperKKL	0.0646	EKF	0.0605	EKF	0.0605	1/8	+7% ±48	GB-EKF	0.00715	+800% ±316
bioreactor (0.841)	HyperKKL	0.0613	EKF-or	0.356	EKF-or	0.356	5/8	**-83%** ±2	GB-EKF-or	0.0214	+187% ±25
duffing (1.7)	HyperKKL	0.0555	UKF	0.00797	UKF	0.00797	0/8	+596% ±17	GB-UKF	0.005	+1011% ±40
lorenz (0.911)	HyperKKL	0.0219	UKF-or	0.157	UKF-or	0.157	0/8	**-86%** ±3	GB-EKF-or	0.0201	+9% ±30
quad_tank (0.607)	HyperKKL	0.0513	KKL-MB	0.434	KKL-MB	0.434	7/8	**-88%** ±2	GB-EKF	0.0135	+279% ±13
vdp (1.11)	HyperKKL	0.02	EKF-or	0.0268	EKF-or	0.0268	0/8	**-25%** ±9	GB-EKF	0.00634	+216% ±43
mismatch20											
battery (0.77)	HyperKKL	0.0569	EKF	0.12	EKF	0.12	1/8	**-53%** ±6	GB-EKF	0.0111	+413% ±82
bioreactor (0.769)	HyperKKL	0.0783	EKF-or	0.401	EKF-or	0.401	6/8	**-80%** ±10	GB-EKF-or	0.0291	+169% ±128
duffing (1.7)	HyperKKL	0.0536	UKF	0.0142	UKF	0.0142	0/8	+278% ±13	GB-UKF	0.00527	+917% ±88
lorenz (0.616)	HyperKKL	0.00772	KKL-MB	0.0248	KKL-MB	0.0248	0/8	**-69%** ±10	NLOX	0.0108	**-28%** ±13
quad_tank (0.587)	HyperKKL	0.0421	KKL-MB	$0.618^{\dagger}$	none	–	8/8	–	GB-UKF	0.0131	+222% ±55
vdp (1.09)	HyperKKL	0.0199	EKF-or	0.0429	EKF-or	0.0429	0/8	**-54%** ±7	GB-EKF	0.00755	+165% ±68
drift30 (post-train)											
battery (0.78)	HyperKKL	0.204	EKF-or	0.161	EKF-or	0.161	1/8	+27% ±4	pend. 0/5		–
bioreactor (0.928)	HyperKKL	4.61	EKF-or	0.255	EKF-or	0.255	6/8	+1708% ±21	pend. 0/5		–
duffing (1.69)	HyperKKL	0.0549	EKF-or	0.0184	EKF-or	0.0184	0/8	+198% ±23	pend. 0/5		–
lorenz (1)	HyperKKL	0.485	UKF-or	0.469	UKF-or	0.469	0/8	+3% ±0	pend. 0/5		–
quad_tank (0.651)	HyperKKL	1.67	KKL-MB	$1.58^{\dagger}$	none	–	8/8	–	pend. 0/5		–
vdp (1.12)	HyperKKL	0.086	EKF-or	0.0711	EKF-or	0.0711	0/8	+21% ±2	pend. 0/5		–
mismatch30											
battery (0.765)	HyperKKL	0.0515	EKF	0.177	EKF	0.177	1/8	**-71%** ±7	GB-EKF	0.0151	+241% ±18
bioreactor (0.708)	HyperKKL	0.0977	EKF-or	0.247	EKF-or	0.247	6/8	**-60%** ±35	NLOX	0.0353	+194% ±317
duffing (1.71)	HyperKKL	0.0561	UKF	0.0216	UKF	0.0216	0/8	+160% ±20	GB-UKF	0.00592	+848% ±22
lorenz (0.614)	HyperKKL	0.007	KKL-MB	0.0142	KKL-MB	0.0142	1/8	**-51%** ±5	NLOX	0.00682	+3% ±38
quad_tank (0.581)	HyperKKL	0.0422	KKL-MB	$0.787^{\dagger}$	none	–	8/8	–	GB-EKF	0.0129	+226% ±24
vdp (1.16)	HyperKKL	0.0205	EKF-or	0.0578	EKF-or	0.0578	0/8	**-65%** ±7	GB-EKF	0.00976	+110% ±52
mismatch50											
battery (0.756)	HyperKKL	0.0503	EKF	0.284	EKF	0.284	1/8	**-82%** ±9	GB-UKF	0.0222	+126% ±47
bioreactor (0.614)	HyperKKL	0.135	UKF-or	0.353	UKF-or	0.353	6/8	**-62%** ±23	NLOX	0.0341	+327% ±474
duffing (1.73)	HyperKKL	0.0549	EKF-or	0.0361	EKF-or	0.0361	0/8	+52% ±2	GB-UKF	0.00802	+587% ±105
lorenz (0.814)	HyperKKL	0.0107	KKL-MB	0.0708	KKL-MB	0.0708	1/8	**-85%** ±17	NLOX	0.00447	+142% ±248
quad_tank (0.6)	HyperKKL	0.0442	KKL-MB	$1.38^{\dagger}$	none	–	8/8	–	GB-EKF	0.0131	+238% ±20
vdp (1.23)	HyperKKL	0.0203	KKL-MB	0.0777	KKL-MB	0.0777	0/8	**-74%** ±3	GB-EKF	0.015	+36% ±33
partial											
battery (0.78)	HyperKKL	0.0744	UKF-or	0.237	UKF-or	0.237	1/8	**-69%** ±12	GB-EKF	0.0191	+295% ±190
bioreactor (0.928)	HyperKKL	0.0753	UKF	$1.63^{\dagger}$	none	–	8/8	–	Koopman KF	0.0271	+179% ±165
duffing (1.69)	HyperKKL	0.0536	UKF-or	0.0845	UKF-or	0.0845	0/8	**-37%** ±14	GB-UKF	0.00775	+593% ±191
lorenz (1)	HyperKKL	0.0308	UKF-or	0.529	UKF-or	0.529	5/8	**-94%** ±4	NLOX	0.0287	+7% ±59
quad_tank (0.651)	HyperKKL	0.0717	EKF-or	0.0202	EKF-or	0.0202	0/8	+255% ±10	GB-EKF	0.0142	+406% ±40
vdp (1.12)	HyperKKL	0.0179	KKL-MB	0.0819	KKL-MB	0.0819	0/8	**-78%** ±3	GB-EKF	0.0257	**-30%** ±15

Table 1: (continued)

scenario / plant (clim.)	BL	NRMSE	BF	NRMSE	BW	NRMSE	fail.	(a) BL vs BW	DD	NRMSE	(b) BL vs DD
realistic											
battery (0.765)	HyperKKL	0.0642	UKF-or	0.29	UKF-or	0.29	1/8	**-78%** ±1	GB-EKF-or	0.0214	+202% ±41
bioreactor (0.708)	HyperKKL	0.0987	EKF-or	2.44†	none	–	8/8	–	NLOX	0.0389	+157% ±106
duffing (1.71)	HyperKKL	0.0997	EKF-or	0.127	EKF-or	0.127	0/8	**-21%** ±2	GB-EKF	0.0144	+591% ±21
lorenz (0.614)	HyperKKL	0.0104	KKL-MB	0.0257	KKL-MB	0.0257	7/8	**-58%** ±11	NLOX	0.0129	**-19%** ±12
quad_tank (0.581)	HyperKKL	0.0441	KKL-MB	1.01†	none	–	8/8	–	GB-EKF	0.017	+160% ±16
vdp (1.16)	HyperKKL	0.0324	KKL-MB	0.125	KKL-MB	0.125	0/8	**-74%** ±1	GB-EKF	0.0388	**-16%** ±5

† FAILED: NRMSE above climatology (in brackets after the plant); fail. = FAILED / present model-based filters.